

Talking Tactics: A Text-Network Analysis of Managers in the Premier League

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This project analyzes how the language used to describe Premier League managers on their Wikipedia pages relates to their tactical identity, historical context, and managerial success. We analyze cleaned biographies and construct a multiplex network based on shared clubs and tactical linguistic similarities. Using TF-IDF, we extract distinctive tactical and leadership vocabulary and evaluate how this linguistic profile, along with tone and sentiment, relates to managerial era, club history, and performance. Our main finding is that tactical language creates strong, meaningful communities that do not align with shared club history or era, and that tactical vocabulary shows a moderate, significant association with managerial success. While sentiment correlates with tactical detail, it does not relate to success or career overlap, suggesting that tactical vocabulary, not tone, is the primary linguistic signal associated with managerial outcomes.

Football | Managers | Terminology | Tactics

Understanding how football managers are represented in written sources reveals key perspectives on their tactical identity, managerial style, and professional reputation. Wikipedia offers a broad collection of biographies that combine career data with descriptions of strategies, achievements, and coaching philosophies. Network analysis provides a way to represent relationships between managers, such as shared clubs, overlapping career paths, or tactical similarities inferred from Wikipedia structures. By combining text features with network relationships, we study whether linguistic similarity reflects deeper structural or historical patterns among managers.

This project analyzes Wikipedia biographies of Premier League managers to explore how linguistic patterns relate to tactical identity, managerial eras, and network structure. Using TF-IDF, tactical terminology lists, sentiment scores, and other lexical measures we identify key words and tactical features that characterize each manager’s biography. These features allow us to compare the language of the profiles across different periods, for example, contrasting traditional “old-school” managers with more modern, analytically oriented “new-school” coaches. We then construct a multiplex network based on career connections and managerial styles to examine whether linguistic similarity corresponds with structural proximity or clustering within the network. A central question is whether tactical vocabulary and tone align with or diverge from patterns created by shared club history.

Finally, we explore whether sentiment, tone, or the use of certain tactical concepts relate to indicators of success on the field, and although we do not assume that language or style directly determines performance, these correlations offer insight into how success is narrated and leadership identities are shaped in public representations. Based on this framework, we evaluate how tactical vocabulary, sentiment, and managerial era relate to performance metrics to determine which linguistic features are most meaningful. By integrating text analysis and network modeling, this project contributes to a deeper understanding of how football managers within the Premier League are portrayed and interconnected within digital knowledge sources.

Significance Statement

This project deep dives into football management by combining text analysis with network science to examine how Premier League managers are connected through both their tactical language and their career histories. Different to most football articles that most focus on statistics and results, our approach shows that the language used to describe managers reveals meaningful patterns in managerial eras, tactical styles, and leadership identity. By constructing a multiplex network we show that managers are connected through distinct yet complementary dimensions. Our results demonstrate that managerial identity extends beyond performance indicators and tactical systems, incorporating both the linguistic narratives surrounding managers and the institutional pathways they follow. By integrating these elements, the project introduces a new perspective on how managerial careers and styles can be interpreted within the broader landscape of contemporary football.

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S.S. wrote the Abstract, Significance Statement, Introduction, Discussion, and Materials and Methods sections.

Á.H.P. performed the majority of the coding, constructed the networks, carried out the community analyses, wrote the corresponding results sections, contributed to the Materials and Methods and overall interpretation, and conducted the full read review to ensure the report formed a coherent and unified text.

E.A.K. implemented and analyzed the correlation experiments, wrote the correlation results section, and contributed to the Materials and Methods.

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Results

A key idea in this project is that Premier League managers are similar to one another in several different ways. Figure 1 presents both layers of the multiplex network in one vertically arranged figure, making it easy to compare them. Each layer represents a different kind of relationship between managers, and the nodes are colored by the year they began managing to reveal any possible temporal patterns.

Figure: Multiplex Network Layers Colored by Start Year

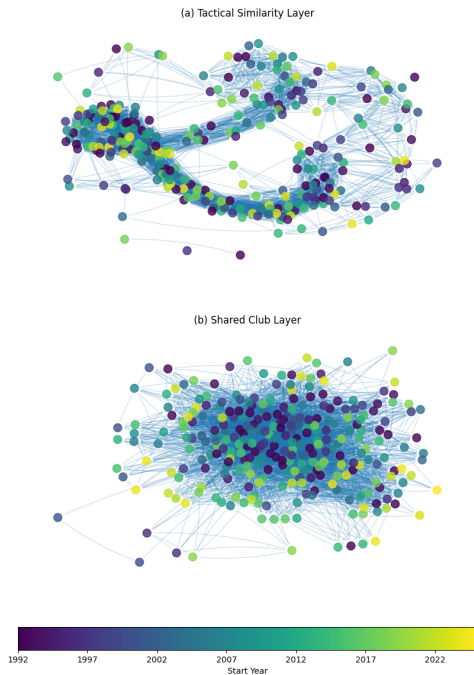


Fig. 1. Two layers of the multiplex network of Premier League managers, with nodes colored by managerial start year. The tactical similarity layer (a) forms a curved, S-like shape due to the force directed layout, highlighting clusters of managers who use similar tactical language in their Wikipedia biographies. The shared club layer (b) is far denser, reflecting the substantial overlap in managerial career paths. Together, the two layers reveal different dimensions of similarity

Tactical Similarity Layer. The tactical similarity layer in Figure 1 a forms a curved, S-shaped pattern, which mainly reflects how the force directed layout arranges the nodes. Early and late managers appear mixed along this curve, suggesting that the structure is shaped far more by shared tactical concepts than by managerial era. Managers whose biographies use similar tactical language naturally cluster together. The start year adds some variation, but it doesn't drive the overall organization of the layer.

Shared Club Layer. The shared club layer, shown in Figure 1 b, is much denser because so many managers have coached at the same clubs. This creates a compact structure where most nodes cluster together regardless of era. Older managers tend to appear near the middle, while newer managers show up around the edges, but overall, the layer remains far less segmented than the tactical one.



Fig. 2. Word cloud of the most frequent football related terms appearing across all managerial biographies. Prominent terms span tactical concepts, positional roles, key achievements, and financial terminology, illustrating the themes that shape managerial narratives.

This suggests that career paths across clubs are highly interconnected and do not form clear groups.

Word Cloud. The tactical word cloud across all eras (Figure 2) visualizes the most frequent football related terms in managers' Wikipedia biographies, forming the core vocabulary of how Premier League managers are described. The distribution of terms confirms that the tactical scoring function successfully captures the key stylistic and contextual elements of managerial descriptions. First, there is a clear emphasis on player development, positional role, and strategic theories, reflected in terms such as "youth", "sporting", "goalkeeper", "formation", and "tackle". Second, the presence of financial and contractual terminology, including "million", "contract", and "legal", highlights the influence of the transfer market and administrative work on modern football and managers' careers. Finally, achievements and competition related words, such as "winner", "trophy", and "award" remain prominent, reflecting their importance in how managerial legacies are narrated. Together, these categories show that biographies combine tactical detail, organizational context, and career achievements, components that later analyses draw upon.

Communities. When we look at the two layers of the multiplex network, it becomes clear that they capture very different kinds of relationships between Premier League managers, as shown through the community detection analysis. The tactical similarity layer is where we see the strongest and most meaningful structure. Its modularity is $M = 0.5894$, which is far higher than what we obtain from randomized networks (null mean ≈ 0.1248 , $SD = 0.0033$). This tells us that the clusters in this layer are meaningful rather than random. In Figure 3, the convex hulls make this separation even clearer: each tactical community forms a distinct and well-defined region, showing that managers who use similar tactical language consistently cluster together. The colors, representing shared-club communities, are scattered across these regions, indicating that shared career paths do not drive tactical structure. The shared-club layer tells a different story. Here, the modularity is

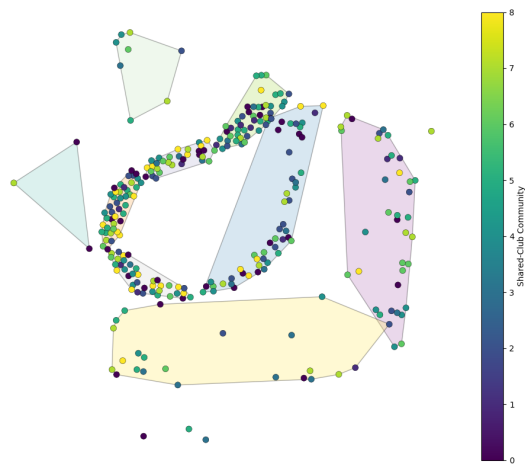


Fig. 3. Convex hulls outlining the Louvain communities detected in the tactical similarity layer, with node colors indicating shared-club community assignments. The figure highlights the strong structural separation between tactical communities and the lack of alignment with shared-club groupings.

much lower $M = 0.2977$, and only slightly higher than the null model (null mean ≈ 0.1140 , $SD = 0.0030$). This reflects the dense interconnectivity produced by overlapping career paths, which prevents meaningful structural separation. Unlike the segmented tactical layer, shared-club communities occupy broad and overlapping regions, showing that managerial movement across clubs does not form coherent structural groupings.

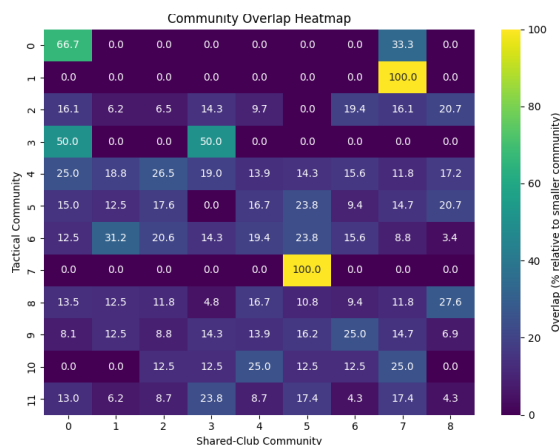


Fig. 4. Illustrates the overlap between tactical communities and shared-club communities. Each cell shows the percentage of managers that belong to both a tactical and shared-club community, measured relative to the smaller community size. Brighter colors indicate higher overlap. The figure highlights that only a few community pairs align strongly, while most show low overlap. This demonstrates that the tactical language used in managers' Wikipedia pages forms clusters that rarely correspond to clusters based on shared career paths, emphasizing that the two layers capture different dimensions of similarity among Premier League managers.

The relationship between the two partitions is summarized in the overlap heatmap in Figure 4. Most tactical communities overlap with multiple shared-club communities, and vice versa. Only a small number of community pairs show substantial alignment, while most exhibit low correspondence. This reinforces that

tactical language and shared career history represent independent dimensions of similarity. This conclusion is further supported by the correlation between the partitions ($r = 0.031$, $p = 0.607$), which is small and not statistically significant. Together, the convex-hull structure, modularity comparison, overlap heatmap, and partition correlation all lead to the same finding: tactical identity, not career path, drives the strongest and most interpretable structure among managers.

Historical results.

Table 1. Correlation Results

Correlation	r	p-value
Sentiment vs Manager Starting Year	0.103	0.0683
Sentiment vs No. of Titles	0.055	0.3543
Sentiment vs Avg Points	0.0728	0.2290
Sentiment vs Total Points	0.103	0.0828
Sentiment vs Avg Points	0.050	0.4006
Sentiment vs Shared Club	-0.0201	0.7402
Sentiment vs Tactical Score	0.352	1.849×10^{-9}
Tactical Score vs Avg Points	0.3542	1.50×10^{-9}
Tactical Score vs No. of Titles	0.3187	6.53×10^{-8}
Shared Club vs Avg Points	-0.2686	0.0624
Share Club vs No. of Titles	-0.0289	0.633

Table 1 summarizes the correlations between sentiment, tactical language, and several managerial performance measures. As can be seen, the sentiment has almost no meaningful correlation to any other factor mentioned in the table. This is consistent with Wikipedia's emphasis on maintaining neutral and unbiased writing on its pages (Wikipedia, 2024. (1)). As a result, sentiment does not reflect managerial era, performance, or the structure of a manager's career. In contrast, the tactical score exhibits clear and statistically robust relationships with several variables.

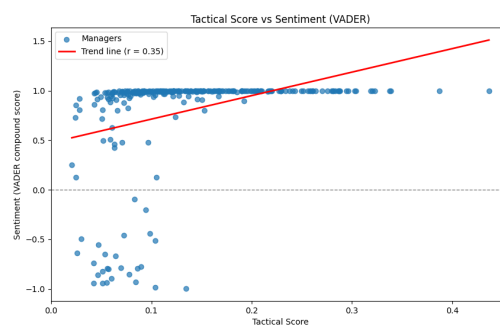


Fig. 5. Scatter plot of tactical score versus VADER sentiment for all managers. A moderate positive correlation ($r = 0.35$) is visible through the fitted trend line, suggesting that pages with higher tactical vocabulary tend to contain slightly more positive evaluative language, despite Wikipedia's overall neutral tone.

First, the tactical score correlates strongly with sentiment (see Figure 5), meaning that biographies containing more tactical terminology tend to have slightly more positive tone. This does not imply that sentiment reflects success, but rather that pages with higher tactical discussion also tend to include more positive descriptive wording. More importantly, the tactical score shows the strongest relationship with managerial

performance: managers whose biographies contain more tactical language tend to have higher average points and more titles. Because tactical detail isn't tied directly to performance, this correlation suggests that tactical vocabulary reflects an underlying aspect of managerial identity, and that successful managers tend to receive more tactically oriented descriptions. The shared club overlap variable, on the other hand, exhibits almost no meaningful correlation with either sentiment or performance. The slight negative correlation with average points is not statistically significant ($p \approx 0.062$), indicating that managers who have coached the same clubs do not systematically resemble one another in success level, tactical language, or narrative tone. This reinforces that shared club history represents a form of structural mobility rather than a meaningful indicator of similarity.

Taken together, these results reveal a clear pattern: sentiment offers almost no useful information because Wikipedia biographies are written in a neutral style, while tactical language stands out as the only linguistic feature meaningfully connected to managerial success, making it a strong indicator of managerial identity. In contrast, shared club overlap shows no real relationship with either linguistic similarity or performance, reinforcing that a manager's career path and their tactical identity operate largely independently.

Discussions

The study explored how Premier League managers are described linguistically and how these descriptions relate to patterns in their careers. By combining text analysis with multiplex network modeling, we identified similarities between managers and uncovered which types of similarity carry meaningful structure.

The tactical similarity layer showed the clearest and strongest structure. Managers cluster based on the tactical vocabulary in their biographies, and this pattern stays consistent across managerial eras. This suggests that tactical identity is shaped more by shared concepts and descriptive styles than by when a manager started coaching. In practice, this reflects how tactical ideas in football evolve slowly: managers from different periods can still be grouped into broader stylistic schools (e.g., possession based vs. defensive minded). In contrast, the shared club layer is much denser and shows little modular structure, largely due to the high managerial turnover in the Premier League. These connections do not form distinct groups, indicating that mobility reflects institutional factors, such as sackings or club cycles, rather than similar coaching styles. Two managers may coach the same club yet belong to very different tactical traditions, which explains the weak alignment between the layers. The correlation analysis further supports this separation. Sentiment has almost no relationship with managerial success or performance, which aligns with Wikipedia's neutral writing style. Its moderate correlation with tactical score simply reflects that pages with more tactical detail tend to use slightly more evaluative or descriptive language. The tactical score, however, shows clear and statistically strong correlations

with performance: managers whose biographies contain more tactical terminology tend to have higher average points and more titles. This does not imply a direct influence, but it does suggest that successful managers are written about with greater tactical depth, either because their approaches draw more attention or because success makes their philosophies easier to describe.

Overall, the study shows that linguistic descriptions capture aspects of managerial identity that structural data alone cannot. Tactical vocabulary, in particular, highlights how managers are perceived and separates successful figures more clearly than shared club history. At the same time, the lack of signal in sentiment and mobility underscores that not all information sources are equally informative. While the findings do not predict results on the pitch, they demonstrate that textual narratives carry real structure, revealing that public descriptions place more emphasis on tactical identity than institutional pathways. This suggests that language offers a valuable lens for understanding how managers are framed within the football world.

Materials and Methods

Explainer notebook. All code behind in the analysis can be found in the [Explainer Notebook](#). There, you can also find a repository that contains information about how the data was downloaded.

Data collection. The dataset combines textual information from Wikipedia with structured Premier League history. Manager biographies were obtained from the category *Premier League managers*, forming the primary corpus for linguistic analysis. To identify football specific terminology, additional Wikipedia biographies listed in [Table 2](#) were collected. These biographies provide domain vocabulary used to detect tactical and positional language. All pages were downloaded as raw text and preprocessed through markup removal, normalization, lemmatization, and custom stopword filtering to exclude names, clubs, nationalities, and non-informative wiki content.

Table 2. Football terminology pages

Wikipedia pages used
Glossary of association football terms
Association football tactics and skills
Formation (association football)
Association football positions
Association football
Category:Association football terminology

Premier League standings from the 1992–2025 seasons were collected from the *English Premier League Standings* dataset on Kaggle. All CSV files were merged into a single table containing clubs, seasons, and finishing positions. Club names were normalized to match those extracted from the manager biographies, enabling links between managers and the Premier League teams they coached. Wikipedia infoboxes were parsed to extract managerial clubs and year ranges, using custom logic to interpret date formats and retain only periods overlapping

Premier League seasons. The resulting set of cleaned biographies, terminology sources, and league records provides the foundation for the linguistic analysis, network construction, and community detection performed in this project.

Network Construction. The multiplex network contains 284 nodes, each representing a Premier League manager. Node attributes include the cleaned Wikipedia biography, a boosted TF-IDF tactical language score, and the manager's estimated Premier League start year. The network consists of two layers. The tactical layer is built from TF-IDF representations of biographies, using football terminology sources to identify tactical vocabulary. After boosting tactical terms and removing non-informative words, a similarity threshold is applied, producing 4,857 edges between managers with comparable tactical language profiles. The shared club layer is constructed from Wikipedia infobox career data. Club names are normalized to Premier League records, and an edge is added whenever two managers coached the same club in Premier League seasons, yielding 5,235 edges.

Multiplex Network. A multiplex network is a network model in which the same set of nodes is connected through multiple distinct layers, with each layer representing a different type of relationship. All layers share the same nodes, but the edges differ depending on the criterion used to link them. This structure makes it possible to analyze how different dimensions of similarity interact within the same system (Wikipedia, 2024 (2)).

TF-IDF. TF-IDF (Term Frequency-Inverse Document Frequency) is a weighting method used to measure how important a word is within a document relative to a larger collection of documents. It increases the weight of terms that appear frequently in a specific text but are rare across the corpus, making it useful for identifying distinctive vocabulary in managerial biographies (Wikipedia, 2024 (3)).

Louvain. The Louvain method is an algorithm for detecting communities in large networks by maximizing modularity, a measure of how strongly a network divides into tightly connected groups. It begins by letting each node form its own community and then repeatedly moves nodes to neighboring communities whenever this improves modularity. Once no further improvements are possible, the algorithm compresses each community into a single node and repeats the process on this reduced network. NetworkX follows this standard two step procedure in its `louvain_communities` implementation, providing an efficient way to uncover meaningful community structure

in networks of many different sizes (NetworkX Developers, 2024 (4)).

Pearson correlation coefficient. The Pearson correlation coefficient measures how strongly two numerical variables move together in a linear way. We used it because it is a clear and widely accepted method for checking whether changes in one variable are connected to changes in another. (5)

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

The p-value for Pearson's r was then found using the t-distribution. (6)

$$t = r\sqrt{\frac{n-2}{1-r^2}}, \quad p = 2(1 - T(|t|, n-2))$$

In our study, Pearson's r was applied to examine how sentiment scores, tactical scores, and historical performance data, such as points, titles, and club history, relate to each other.

Sentiment Analysis. VADER is a rule based sentiment analysis tool designed for short and informal text. It uses a predefined lexicon and heuristic rules to assign positive, negative, neutral, and compound sentiment scores to a sentence, making it effective for analyzing brief descriptive texts such as manager Wikipedia biographies (GeeksforGeeks, 2024 (7)).

Limitations

A key limitation concerns assigning managerial success to calendar years instead of Premier League seasons. Because seasons span two calendar years, mid-season managerial changes may lead to minor misattributions of titles or points. This effect is limited, as sackings typically occur during poor performances and rarely influence title outcomes. A second limitation is the use of Wikipedia biographies, where article length, tactical detail, and narrative framing vary by editor activity rather than managerial reality. This may inflate tactical vocabulary for well known or successful managers. Methodologically, the tactical score depends on our chosen terminology lists and boosting scheme, introducing some subjectivity into what counts as "tactical language." Different weighting choices could shift edge strengths in the tactical layer. Finally, the similarity threshold used to construct this layer shapes community detection results; although null-model comparisons indicate robustness, community boundaries should still be interpreted in light of these modeling decisions.

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